

ITA0303 - Mobile Computing

CO1 - Assessment Tools

01. Full-duplex voice channels.

Given

Total spectrum = 45 MHz

Simple channel bandwidth = 25 kHz

No. of Simplex channels = $\frac{\text{Total bandwidth}}{\text{channel bandwidth}}$

No. of Full duplex channels = $\frac{\text{Simplex channels}}{2}$

Convert MHz to kHz:

45 MHz = 45,000 kHz

Simple channels:

$$45000 \div 25 = 1800$$

Full-duplex channels:

$$1800 \div 2 = 900 //$$

2. channels per cell

Given

Total channels = 560

cluster size = 7

channels per cell = $\frac{\text{Total channels}}{\text{cluster size}}$

$$\frac{560}{7} = 80 //$$

3) GSM voice channels.

Given

No. of carriers = 100

Time slots per carrier = 8

Total voice channels = carriers $\times$ Time slots

$$100 \times 8 = 800 //$$

4) Power control ratio

Given:

Base station power = 80W

Handset power = 1W

$$\text{Power ratio (dB)} = 10 \log_{10} (P_1/P_2)$$

$$= 10 \log_{10} (80/4)$$

$$= 10 \log_{10} (20)$$

$$= 10 \times 1.301$$

$$= 13.01 \text{ dB}$$

$$\text{Ans: } \approx 13 \text{ dB} //$$

5) Carrier-to-Interference Ratio

Given

$$\text{Signal} = -60 \text{ dBm}$$

$$\text{Interference} = -92 \text{ dBm}$$

$$\text{C/I} = \text{Signal} - \text{Interference}$$

$$= (-60) - (-92)$$

$$= 32 \text{ dB} //$$

6) Channels per cell

Given

$$\text{Spectrum} = 42 \text{ MHz}$$

$$\text{Channel bandwidth} = 300 \text{ kHz}$$

$$\text{Cluster size} = 7$$

$$42 \text{ MHz} = 42,000 \text{ kHz}$$

$$\text{Total channels} = 42,000 \div 300 = 140 //$$

$$\text{Channels per cell} = 140 \div 7 = 20 //$$

20 channels per cell.

7) Received power after doubling distance

Given

$$\text{Initial power} = 64 \text{ W}$$

Distance doubled

$$\text{Path loss exponent} = 4$$

$$\text{New power} = \text{old power} \div (2^n)$$

$$= 64 \div 16$$

8) Increase transmit power.

Given

cell radius doubled

Inverse square law ($n=2$)

$$\text{power increase} = (2^2) = 4$$

Transmit power must increase by a factor of 4

9) Total channels in cluster.

Given

$$\text{cluster size} = 4$$

$$\text{channels per cell} = 15$$

$$\text{Total channels} = \text{cluster size} \times \text{channels per cell}$$

$$= 4 \times 15 = 60$$

60 channels //

10) Doppler shift

Given

$$\text{Speed} = 100 \text{ km/h}$$

$$\text{Frequency} = 900 \text{ MHz}$$

Convert speed

$$100 \text{ km/h} = 100 \times 1000 / 3600 = 27.78 \text{ m/s}$$

$$f_d = \frac{v f}{c}$$

$$v = 27.78 \text{ m/s}$$

$$f = 900 \times 10^6 \text{ Hz}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$f_d = \frac{27.78 \times 900 \times 10^6}{3 \times 10^8} = 83.34 \text{ Hz} //$$

11) Duration of one TDMA time slot

Given

$$\text{Frame duration} = 5 \text{ ms}$$

$$\text{No. of slots} = 8$$

$$\text{Slot duration} = \frac{\text{Frame duration}}{\text{Number of slots}}$$

$$= \frac{5}{8} = 0.625 \text{ ms} \quad (625 \mu\text{s}) //$$

12) Number of GSM carrier frequencies.

Given

Total bandwidth = 24 MHz

Carrier Spacing = 200 kHz

$$24 \text{ MHz} = 24,000 \text{ kHz}$$

$$\text{Carriers} = \frac{\text{Total bandwidth}}{\text{Carrier Spacing}}$$

$$= \frac{24,000}{200} = 120$$

120 carrier frequencies //

13) Received power after path loss =

Given

Transmit power = 8 W

path loss = 33 dB

Convert 8 W to dBm

$$P(\text{dBm}) = 10 \log_{10}(8000) = 39.03 \text{ dBm}$$

Received power:

$$P_r = P_t - \text{path loss}$$
$$= 39.03 - 33 = 6.03 \text{ dBm} //$$

14) Free Space path loss.

Given

frequency = 1900 MHz

Distance = 2.5 km

$$\text{FSPL} = 32.44 + 20 \log_{10}(f_{\text{MHz}}) + 20 \log_{10}(d_{\text{km}})$$

$$\text{FSPL} = 32.44 + 20 \log_{10}(1900) + 20 \log_{10}(2.5)$$

$$= 32.44 + 65.58 + 7.96$$

$$= 105.98 \text{ dB}$$

$$\approx 106 \text{ dB} //$$

15) clutch call - handling capacity of system

Given

No. of cells = 6

calls per cell = 250

Total capacity = $6 \times 250 = 1500$

1500 simultaneous calls.

16) Maximum active calls.

Given

New call requests = 4000 calls/sec.

Average call duration = 3 minutes.

3 minutes = 180 seconds.

Active calls = $\lambda \times T$

= $4000 \times 180 = 720,000$ active calls //

17) Total bandwidth Consumed

Given

No. of channels = 150

channel bandwidth = 25 kHz

$B = N \times BW$

= $150 \times 25 = 3750$ kHz

= 3.75 MHz //

18) Bandwidth wasted by guard bands.

Given

Guard band = 40 kHz

No. of guard bands = 30

wasted bandwidth = 30×40

= 1200 kHz

= 1.2 MHz //

19) Number of handoffs in one hour.

Given

Speed = 90 km/h

Cell radius = 1.5 km

Cell diameter = $2R = 2(1.5) = 3$ km //

Distance travelled in 1 hour: 90 km

Hand offs = $\frac{90}{3} = 30$

Approximately 30 handoffs

20y Total user capacity

Given

Regional Centers = 8

Base stations per region = 40

Users per base station = 250

Total users = $8 \times 40 \times 250 = 80,000$ users //

21y Number of cells required.

Given

Total area = 400 km²

Cell radius = 400 m = 0.4 km

Approximate cell area = R^2

Cell Area = R^2

Cell area = $(0.4)^2 = 0.16$ km²

No. of cells = $\frac{400}{0.16} = 2500$ cells //

22y Number of FDMA channels.

Given

Total Spectrum = 6 MHz

channel bandwidth = 30 kHz

6 MHz = 6000 kHz

channels = $\frac{\text{Total bandwidth}}{\text{channel bandwidth}}$

= $\frac{6000}{30} = 200$ channels //

23y Offered traffic (Erlangs)

Subscribers = 1200

calls per subscriber = 4 calls / hour

Average call duration = 2.5 minutes

$$2.5/60 = 0.04167 \text{ hour}$$

$$A = N \times \lambda \times h$$

$$= 600 \times 4 \times 0.04167$$

$$= 200 \text{ Erlangs} //$$

25) Maximum Simultaneous users.

$$\text{Total bandwidth} = 40 \text{ MHz}$$

$$\text{Bandwidth per user} = 400 \text{ kHz}$$

$$40 \text{ MHz} = 40,000 \text{ kHz}$$

$$\text{Users} = \frac{40,000}{400} = 100 \text{ users} //$$

26) Total network user capacity.

$$\text{users per base station} = 600$$

$$\text{No. of base stations} = 1200$$

$$\text{Total users} = 600 \times 1200 = 720,000 \text{ users} //$$

26) Paging rounds

Given

$$\text{Total cells} = 200$$

$$\text{cells searched per round} = 4$$

$$\text{paging rounds} = \frac{200}{4} = 50 \text{ paging rounds} //$$

27) fraction of traffic blocked.

Given

$$\text{Maximum Simultaneous Connections} = 100,000$$

$$\text{call attempts} = 40,000/\text{min}$$

$$\text{Average duration} = 3 \text{ min}$$

Offered traffic:

$$40,000 \times 3 = 120,000$$

$$\text{Blocked calls} : 120,000 - 100,000 = 20,000$$

$$\text{Blocking fraction} : \frac{20,000}{120,000} = 0.1667$$

$$\text{Percentage} : 0.1667 \times 100 = 16.67\% \text{ blocked} //$$

28)

Roaming Subscribers.
Given

$$\text{Total Subscribers} = 600,000$$

$$\text{Roaming percentage} = 15\%$$

$$\text{Roaming users} = 600,000 \times 0.15 = 90,000$$

29) Switching Center. utilization

Given

$$\text{Capacity} = 12,000 \text{ calls/sec}$$

$$\text{Current load} = 9,000 \text{ calls/sec}$$

$$\text{utilization} = \frac{9000}{12000} \times 100 = 75\% \text{ utilization}$$

30) channels per cell

Given

$$\text{Total Spectrum} = 48 \text{ MHz}$$

$$\text{channel bandwidth} = 200 \text{ kHz}$$

$$\text{Reuse Factor} = 4$$

$$48 \text{ MHz} = 48,000 \text{ kHz}$$

$$= \frac{48,000}{200} = 240$$

$$\text{channels per cell} = \frac{240}{4} = 60 \text{ channels per cell}$$

31) Number of GSM channels.

Given

$$\text{Total Spectrum} = 30 \text{ MHz}$$

$$\text{channel spacing} = 200 \text{ kHz}$$

$$30 \text{ MHz} = 30,000 \text{ kHz}$$

$$\text{channels} = \frac{\text{Total bandwidth}}{\text{channel spacing}}$$

$$= \frac{30,000}{200} = 150 \text{ channels}$$

32) Successful handoffs

Given

$$\text{Handoff attempts} = 2500$$

$$\text{Success rate} = 99\%$$

$$\text{Successful handoffs} = \frac{99}{100} \times 2500$$

$$= 2475 \text{ Successful handoffs}$$

33) Received power at the base station

Given

$$\text{Transmit power} = 12 \text{ W}$$

$$\text{path loss} = 45 \text{ dB}$$

$$P_t (\text{dBm}) = 10 \log_{10} (121000) = 40.79 \text{ dBm}$$

$$P_r = P_t - \text{path loss} = 40.79 - 45$$

$$= -4.21 \text{ dBm} \approx -4.2 \text{ dBm}$$

34) cluster user capacity

Given

$$\text{cluster size} = 7 \text{ cells}$$

$$\text{users per cell} = 900$$

$$\text{Total users} = 7 \times 900$$

$$= 6300 \text{ users}$$

35) TDMA slot duration?

Given

$$\text{frame duration} = 4.8 \text{ ms}$$

$$\text{Slots} = 8$$

$$\text{slot duration} = \frac{\text{frame duration}}{\text{slots}}$$

$$= \frac{4.8}{8} = 0.6 \text{ ms} \quad (600 \mu\text{s}) //$$

36) Doppler shift

Given

$$\text{Speed} = 150 \text{ km/h}$$

$$\text{frequency} = 1.8 \text{ GHz}$$

$$150 \times \frac{1000}{3600} = 41.67 \text{ m/s}$$

$$f_d = \frac{v_f}{c}$$

$$f_d = \frac{41.67 \times 1.8 \times 10^9}{3 \times 10^8} = 250 \text{ Hz}$$

37) Share of offered load, blocked

Given

System capacity = 250,000

Call arrival rate = 10,000 calls/min

Avg call duration = 4 min

Offered traffic

$$= 10,000 \times 4 = 280,000$$

Blocked calls:

$$280,000 - 250,000 = 30,000$$

$$\text{Blocking percentage} = \frac{30,000}{280,000} \times 100$$

$$= 10.71\% \text{ blocked}$$

38) cluster user Capacity

Given

cluster size = 9 cells

users per cell = 600

Total users = $9 \times 600 = 5400$ users

39) Calls dropped per Second

Given

calls processed = 60,000 calls/sec

Drop rate = 0.4%

$$\text{Dropped calls} = \frac{0.4}{100} \times 60,000 = 240 \text{ calls/sec.}$$

40) Path loss

Given

Transmit power = 43 dBm

Received power = -87 dBm

Path loss = $P_t - P_r$

$$: U_3 = (-87) = 130 \text{ dB} //$$

u1) Total geographic area covered

Given

Area covered by one base station = 2.5 km^2

No. of base stations = 150

Total Area = Area per station $\times$ No. of stations

$$= 2.5 \times 150 = 375 \text{ km}^2 //$$

u2) channels per cell

Given

Total Spectrum = 48 MHz

channel bandwidth = 200 kHz

frequency reuse factor = 6

$48 \text{ MHz} = 48,000 \text{ kHz}$

$$\text{Total channels} = \frac{48,000}{200} = 240$$

$$\text{channels per cell} = \frac{240}{6} = 40 //$$

u3) Paging Cycles

Given processed per paging cycle = 6000

Total subscribers = 150,000

$$\text{paging cycles} = \frac{\text{Total subscribers}}{\text{users per cycle}}$$

$$= \frac{150,000}{6000}$$

$$= 25 //$$

u4) Number of handoffs during a call

Given

Call duration = 9 minutes

Handoff every 3 minutes

Handoffs = $\frac{\text{Call duration}}{\text{Handoff Interval}}$

$$= \frac{9}{3} = 3 //$$

Q15) Signal-to-Noise Ratio (SNR)

Given

Signal power = -85 dBm

Noise power = -115 dBm

$$SNR = \text{Signal} - \text{Noise}$$

$$= -85 - (-115) = 30 \text{ dB}$$

Q16) Minimum bandwidth required

Given

No. of users = 250

Bandwidth per user = 150 kbps

$$\text{Total bandwidth} = 250 \times 150$$

$$= 37,500 \text{ kbps}$$

$$= 37.5 \text{ Mbps}$$

Q17) Total traffic (Erlangs)

Given

Subscribers = 1500

calls per subscriber = 4 calls/hour

Average call duration = 2.5 minutes

$$2.5 \text{ min} = \frac{2.5}{60} = 0.04167 \text{ hour}$$

$$A = N \times \lambda \times h$$

$$= 1500 \times 4 \times 0.04167$$

$$= 250 \text{ Erlangs}$$

Q18) Bandwidth for Control channels

Given

No. of Control channels = 250

Bandwidth per channel = 12 kHz

$$\text{Total bandwidth} = 250 \times 12$$

$$= 3000 \text{ kHz}$$

$$= 3 \text{ MHz}$$

495 Total network throughput

Simultaneous connections : 1500

Data rate per Connection : 600 kbps

$$\text{Through put} = 1500 \times 600$$

$$= 900,000 \text{ kbps}$$

$$= 900 \text{ mbps (0.9 Gbps)}$$

505 Number of towers required

Given

Coverage radius : 4 km

Area to cover : 350 km²

Coverage area of one tower.

$$A = \pi R^2$$

Area covered by one tower :

$$= 3.14 \times 4^2$$

$$= 3.14 \times 16 = 50.24 \text{ km}^2$$

$$\text{No. of towers} = \frac{350}{50.24} = 6.97$$

Since a fraction of a tower is not possible,

Round up : = 7 //